SIT1001 - Week 3 Unit 2 Part 2 (1 hours)

2.4 Properties of the random variables

Two main properties will be studied in depth:

a. Mathematical expectation – Mean and Variance
b. Moment generating function (MGF)

2.4.1 Mathematical Expectation

An extremely important concept in summarizing important characteristics distributions of probability.

Definition 2.4 (Def 3.2-1) *If f(x) is the p.m.f of the random variable X of the discrete type with space S and if the summation*

$$\sum_{x\in S} u(x)f(x), \qquad \textit{which is sometimes written} \qquad \sum_{S} u(x)f(x),$$

exists, then the sum is called the ***mathematical expectation*** *or* ***the expected value*** *of the function u(X), and it is donated by E[u(X)]. That is,*

$$E[u(X)] = \sum_{x\in S} u(x)f(x)$$

Note: When X **is continuous variable**, then $E[u(X)] = \int_S u(x)f(x)dx$.

Note: (i) Expected value of a discrete variable X: $E[X] = \sum_{x\in S} x f(x)$

(ii) Expected value of a discrete variable $u(x) = (X-\mu)^2$: $E\left[(X-\mu)^2\right] = \sum_{x\in S}(X-\mu)^2 f(x)$

The earlier is referred as the **mean**, μ, and the latter as the **variance**, σ^2. Similarly for a continuous variable,

$$\mu = E[X] = \int_S x f(x)dx \text{ and } \sigma^2 = E\left[(X-\mu)^2\right] = \int_S (X-\mu)^2 f(x)dx.$$

Note: (iii) $\sigma^2 = E\left[(X-\mu)^2\right] = E\left[X^2 - 2\mu X + \mu^2\right] = E\left[X^2\right] - 2\mu E[X] + \mu^2 = E\left[X^2\right] - \mu^2$.

Example 2.10 (Ex 3.2-2 H&T) Let the random variable X have the p.m.f. $f(x)=\frac{1}{3},\ x\in S,$ where $S=\{-1,0;1\}$. Let $u(X)=X^2$. Find $E[X]$ and $E[u(X)]$.

Solution:

Example 2.11 (Ex 4.1-8 H&T) Let X have the p.d.f $f(x)=e^{-x-1},-1<x<\infty$. Find μ and σ^2.

Theorem 2.1 (Thm 3.2-1, H&T) When it exists, mathematical expectation satisfies the following properties:

a) If c is a constant, $E(c) = c$.

b) If c is a constant and u is a function, $E[cu(X)] = cE[u(X)]$

c) If c_1 and c_2 are constant and u_1 and u_2 are functions then,

$$E[c_1u_1(X) + c_2u\ (X)] = c_1E[u_1(X)] + c_2E[u_2(X)]$$

Proof: (a) $E(c) = \sum_{x\in S} cf(x) = c\sum_{x\in S} f(x) = c$ because $\sum_{x\in S} f(x) = 1$

(b) $E[cu(x)] = \sum_{x\in S} cu(x)f(x) = c\sum_{x\in S} u(x)f(x) = cE[u(X)]$

(c) $$\begin{aligned} E[c_1u_1(X) + c_2u_2(X)] &= \sum_{x\in S}[c_1u_1(x) + c_2u_2(x)]f(x) \\ &= \sum_{x\in S} c_1u_1(x)f(x) + \sum_{x\in S} c_2u_2f(x) \\ &= c_1E[u_1(X)] + c_2E[u_2(X)] \end{aligned}$$

Note: Property (c) can be extended to more than two terms by mathematical induction:

$$E\left[\sum_{i=1}^{k} c_iu_i(X)\right] = \sum_{i=1}^{k} c_iE[u_i(X)].$$

Note: **These properties are true for both discrete and continuous variables (the proof above can be shown to be true for the continuous case)**.

Example 2.12 (Ex 3.2-3 H&G) Let X have a p.m.f $f(x) = \frac{x}{10}$, $x = 1,2,3,4$. Find $E(X), E(X^2), E[X(5-X)]$.

Solution:

Example 2.16 (Ex 3.2-6 H&T) Let the p.m.f of X be defined by $f(x) = x/6, x = 1,2,3$. What the mean, variance and standard deviation of X?

Solution:

Example 2.17 (Ex 3.2-7 H&T) The mean of X, which has a uniform distribution on the first m positive integers, is given by $\mu = E(X) = \sum_{x=1}^{m} x\left(\frac{1}{m}\right) = \frac{1}{m}\sum_{x=1}^{m} x$. Find the variance, σ^2 of X.

Solution:

Generalization of the mathematical expectation

Let r be a positive integer.

(a) If $E(X^r) = \sum_{x\in R} x^r f(x)$ is finite, it is called the ***r*-th moment** of the distribution about the origin.

(b) In addition, the expectation $E\left[(X-b)^r\right] = \sum_{x\in R}(x-b)^r f(x)$ is called the ***r*-th moment** of the distribution about b.

(c) For positive integer r, $E\left[(X)_r\right] = E\left[X(X-1)(X-2)...(X-r+1)\right]$ is called r th **factorial moment**. Note that second factorial moment is the difference of second and first moments: $E\left[X(X-1)\right] = E(X^2) - E(X)$.

(d) Variance can be computed from second factorial moment and sometimes simplies the calculation. First, find the values of $E(X)$ and $E[(X)(X-1)]$, Then

$$\sigma^2 = E\left[X(X-1)\right] + E(X) - \left[E(X)\right]^2,$$

Since, by using **distributive property of *E*,** this becomes

$$\sigma^2 = E(X^2) - E(X) + E(X) - \left[E(X)\right]^2 = E(X^2) - \mu^2.$$